MELODIO — Complete Platform Audit

Everything Built As Of March 17, 2026

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1. PLATFORM OVERVIEW

Name: Melodio (formerly ECHO) **Repository:** github.com/GKOffice/echo-music-company **Stack:** Next.js 14 + FastAPI + PostgreSQL 15 + Redis 8 **Total Commits:** 22 **Total Files:** 491+ **Lines of Code:** 34,000+ **Blueprint:** v18.0 — 22,147 lines, 28 sections (A–AB)

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2. CODEBASE STRUCTURE

```
/Users/bm-007/projects/echo/
├── apps/
│   ├── api/          — FastAPI backend (Python)
│   ├── web/          — Next.js 14 frontend (TypeScript)
│   └── agents/       — 24 AI agents (Python)
├── packages/
│   ├── db/schema.sql — Database schema (21 tables)
│   ├── config/       — Shared constants & config
│   └── shared/       — TypeScript interfaces
├── infra/
│   └── scripts/      — Setup scripts
└── docker-compose.yml
```

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3. AI AGENTS (24 Total)

Each agent is a Python class with Redis pub/sub messaging, database access, and optional Claude AI brain.

#	Agent	File	Purpose
1	CEO Agent	ceo.py	

#	Agent	File	Purpose
			Signing approvals, budget governance, crisis handling, daily briefing
2	A&R Agent	ar.py	6-factor scoring algorithm, demo processing, signing recommendations
3	Production Agent	production.py	Track production oversight, quality control
4	Distribution Agent	distribution.py	Release pipeline, platform uploads, editorial pitching
5	Marketing Agent	marketing.py	Campaign management, 80% marketing rule, budget optimization
6	Social Media Agent	social.py	Content scheduling, engagement tracking
7	Finance Agent	finance.py	Royalty calculations, expense tracking, tax compliance
8	Legal Agent	legal.py	Worldwide compliance (GDPR, Berne, ICC, FATF, DSA, 60+ PROs)
9	Analytics Agent	analytics.py	Data intelligence, anomaly detection, performance dashboards
10	Creative Agent	creative.py	Artwork, branding, visual identity
11	Sync Agent	sync.py	TV/film licensing, sync deal management
12	Artist Dev Agent	artist_dev.py	Career development, milestone tracking
13	PR Agent	pr.py	Press releases, media outreach
14	Comms Agent	comms.py	All artist communication (exclusive channel)
15	QC Agent	qc.py	Quality assurance, content review
16	Infrastructure Agent	infrastructure.py	

#	Agent	File	Purpose
			System health, monitoring, disaster recovery
17	Demo Intake Agent	intake.py	Submission processing, initial screening
18	Merch Agent	merch.py	Digital merchandise, store management
19	YouTube Agent	youtube.py	Video content, channel optimization
20	Producer Hub Agent	hub.py	Beat marketplace, producer management
21	Vault Agent	vault.py	Points pricing engine, exchange, quarterly payouts
22	Deal Room Agent	deal_room.py	B2B rights trading marketplace
23	Fan Intelligence Agent	fan_intelligence.py	Personalized artist discovery, momentum scoring
24	Artist Intelligence	(via API router)	Cross-platform search, AI analysis, predictive signals

Agent Infrastructure: - `base_agent.py` — BaseAgent class with database, Redis, and Claude integration - `bus.py` — Redis pub/sub + streams message bus - `main.py` — Full 21-agent orchestrator (boots all agents)

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4. WEB PAGES (27 Total)

Page	URL	Description
Landing Page	`/`	Hero section, platform stats, featured artists
Discover	`/discover`	Artist discovery with search & filters
Points Store	`/points`	Browse and purchase Melodio Points
Point Detail	`/points/[id]`	Individual point drop page
Digital Store	`/store`	Digital merchandise marketplace
Deal Room	`/dealroom`	B2B rights trading marketplace
Deal Detail		Individual deal listing

Page	URL	Description
	`/dealroom/[id]`	
Create Deal	`/dealroom/create`	List a new deal
Submit Demo	`/submit`	Artist demo submission form
How It Works	`/how-it-works`	Full explainer with FAQ accordion
Artist Dashboard	`/dashboard`	Artist hub — songs, revenue, submissions
Digital Store Mgmt	`/dashboard/digital-store`	Artist merch management
Fan Dashboard	`/fan/dashboard`	Portfolio, royalties earned, holdings
Artist Profile	`/artist/[slug]`	Public artist profile page
Songwriter Hub	`/songwriters`	Revenue sources, tiers, co-write opportunities
Songwriter Register	`/songwriters/register`	Song registration wizard
Songwriter Dashboard	`/songwriters/dashboard`	Revenue breakdown, collection status
Co-Write Marketplace	`/songwriters/cowrite`	Find co-writing opportunities
Ambassador	`/ambassador`	Referral tracking dashboard
Leaderboards	`/leaderboards`	Top Artists / Top Supporters / Rising Stars
Transparency	`/transparency`	Platform-wide public stats
Transactions	`/transactions`	Full transaction history
Notifications	`/notifications`	Notification feed
Notification Settings	`/settings/notifications`	Toggle preferences
Intelligence Search	`/intelligence`	**NEW — Search any artist, AI analysis**
Intelligence Report	`/intelligence/[name]`	**NEW — Full AI report with 10 predictive signals**

Page	URL	Description
Login / Signup	`/auth/login`, `/auth/signup`	Authentication pages

UI Features: - Dark/Light Mode toggle - PWA enabled (installable app — manifest.json + service worker) - Mobile responsive (hamburger nav, responsive grids) - Loading skeletons on all data-fetching pages - Melodio branding throughout (#0a0a0f bg, #8b5cf6 purple, #10b981 green)

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5. API ENDPOINTS (127+ Total)

Routers (15):

Router	Prefix	Endpoints
Auth	`/api/v1/auth`	Login, signup, refresh, profile
Artists	`/api/v1/artists`	CRUD, search, roster management
Releases	`/api/v1/releases`	Create, list, track management
Points	`/api/v1/points`	Point drops, purchases, portfolio
Agents	`/api/v1/agents`	Task submission, status, messages
Hub	`/api/v1/hub`	Producer marketplace, beat uploads
Finance	`/api/v1/finance`	Royalties, expenses, payouts
Legal	`/api/v1/legal`	Contracts, DMCA, compliance
Analytics	`/api/v1/analytics`	Dashboards, anomaly scans
Distribution	`/api/v1/distribution`	Release pipeline, platform status
Deal Room	`/api/v1/deal-room`	Listings, offers, auctions, catalog
Songwriters	`/api/v1/songwriters`	Registration, dashboard, co-writes
Digital Merch	`/api/v1/digital-merch`	Products, purchases, bundles

Router	Prefix	Endpoints
Payments	`/api/v1/payments`	Stripe checkout, webhooks
Fan Intelligence	`/api/v1/fan-intelligence`	Momentum scoring, trending, taste
Artist Intelligence	`/api/v1/intelligence`	**NEW — Search, report, compare**

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6. DATABASE (34 Tables)

Table	Purpose
users	User accounts, auth
sessions	Login sessions
artists	Artist profiles, scores, status
producers	Producer profiles
releases	Album/single releases
tracks	Individual tracks
echo_points	Point ownership records
royalties	Royalty calculations & payouts
contracts	Legal agreements
agent_tasks	AI agent task queue
agent_messages	Inter-agent communication
submissions	Demo submissions
contacts	Artist contacts
audit_log	All system actions
hub_beats	Producer beat marketplace
expenses	Financial expenses
copyright_registrations	Copyright filings
dmca_requests	Takedown requests
point_drops	Point release events

Table	Purpose
exchange_orders	Points exchange order book
exchange_trades	Completed exchange trades
point_payouts	Quarterly payout records
track_price_history	Dynamic pricing history
campaigns	Marketing campaigns
communications	Artist communications log
analytics_snapshots	Performance snapshots
deal_listings	Deal Room listings
deal_offers	Deal Room offers
deal_messages	Deal Room messaging
deals	Completed deals
distribution_submissions	Platform submissions
release_platforms	Release × platform status
external_catalog	Verified external catalog
merch_products	Digital merchandise
digital_products	Downloadable products
digital_purchases	Purchase records
digital_bundle_items	Bundle components

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7. ARTIST INTELLIGENCE (NEW FEATURE)

What It Does

Search any artist by name → system pulls data from multiple platforms → Claude AI generates a comprehensive predictive analysis report.

Data Sources (6 Adapters):

1. **Spotify** — Followers, monthly listeners, popularity, top tracks, genres (requires API key)
2. **YouTube** — Subscribers, total views, recent videos (requires API key)
3. **Genius** — Song count, pageviews, top songs (free API)

4. **MusicBrainz** — Discography, release history, collaborators, labels (free API)
5. **Bandsintown** — Upcoming shows, past events, venue data (free API)
6. **Wikipedia** — Career summary, awards, achievements (free API)

Melodio Score (0-100):

- Music Quality Signal (35%) — release consistency, production collaborations
- Social Momentum (20%) — follower growth velocity, cross-platform presence
- Commercial Traction (25%) — streaming numbers, playlist potential
- Brand Strength (10%) — visual identity, narrative clarity
- Market Timing (10%) — genre trend alignment, competitive landscape

10 Predictive Signals (Innovation):

#	Signal	What It Measures
1	🔗 Collaboration Network	Who they work with — rising producers = leading indicator
2	🚀 Platform Velocity	Growth SPEED not just size — 1K→10K in 3mo > sitting at 500K
3	🎵 Genre Timing	Is their genre trending up or saturated?
4	📱 Content-to-Music Ratio	Non-music content creators grow faster
5	🎤 Live Performance Trajectory	Venue size progression over time
6	🎬 Sync Readiness	TV/film licensing potential based on style
7	💎 Fanbase Quality	Engagement rate > follower count
8	📅 Release Cadence	Consistent releasers (6-8 weeks) grow faster
9	🔄 Cross-Platform Correlation	High correlation = organic. Low = bots/playlist-only
10	💥 Breakout Probability	AI-generated % chance of 10x growth in 12 months

Report Includes:

- Melodio Score gauge with 5-dimension breakdown
- Sign recommendation with confidence percentage
- 10 predictive signal cards with trend arrows (↑↓→)
- Key strengths & key risks
- Collaboration network visualization
- AI Executive Summary (3-4 paragraph deep dive)

- Platform stats grid
- Discography timeline
- Upcoming shows
- Wikipedia background

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8. INTEGRATIONS CONFIGURED

Service	Status	Purpose
Stripe	✅ Test keys active	Payments, checkout, Connect
Anthropic (Claude)	✅ API key active	AI agent brains, analysis
OpenAI	✅ API key active	Alternative AI
PostgreSQL 15	✅ Running locally	Primary database
Redis 8	✅ Running locally	Caching, agent messaging
Spotify	⬜ Key needed	Artist data enrichment
YouTube	⬜ Key needed	Video/channel analytics
SoundCloud	⬜ Key needed	Audio platform data
Chartmetric	⬜ Key needed	Industry analytics
AWS S3	⬜ Key needed	File storage
SendGrid	⬜ Key needed	Email notifications
Instagram	⬜ Key needed	Social data
TikTok	⬜ Key needed	Social data
Twitter	⬜ Key needed	Social data
Persona	⬜ Key needed	KYC verification

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9. BLUEPRINT v18.0 (22,147 lines)

Section	Content
Part A	Build Plan
Part B	Company Overview

Section	Content
Part C	21 Agent Definitions
Part D	Shared Systems
Part E	Multi-Agent Workflows (12)
Part F	Flexible Deals — No Long-Term Contracts
Part G	ECHO Publishing (95% to writer)
Part H	Royalty Collection Infrastructure
Part I	Open Platform ($29-499/mo)
Part J	Label Partnerships ($99-799/mo)
Part K	Points — Master & Publishing
Part L	Ambassador Program (invite-only)
Part M	Legal — Deal Structure & Splits
Part N	Backend Architecture, Auth & Security
Part O	No-Code Build Guide
Part P	Agent Scalability
Part Q	Configuration & 80+ Thresholds
Part R	Agent Communication Protocol
Part S	Technology Stack & Costs
Part T	Tool Audit (42 gaps identified)
Part U	Artist Experience (17-chapter journey)
Part V	Ranking System (leaderboards, 60+ badges)
Part W	Social Proof & Public Stats
Part X	Innovation Roadmap (24 features)
Part Y	Policies & Compliance (18 sections)
Part Z	Agent Ideas (11 specs)
Part AA	5 New Workflows
Part AB	21-Agent Review (47 findings)

10. SUPPORTING DOCUMENTS

Document	Location
Blueprint v18 (MD)	music-label/FULL-BLUEPRINT.md
Blueprint v18 (PDF)	music-label/FULL-BLUEPRINT.pdf (4.7MB)
21 Agent Specs	music-label/agents/ (01-21)
12 Workflows	music-label/workflows/
Thresholds Config	music-label/config/thresholds.yaml
Artist Experience	music-label/docs/artist-experience.md
Ranking System	music-label/docs/ranking-system.md
Social Proof Stats	music-label/docs/social-proof-stats.md
Legal Structure	music-label/docs/legal-deal-structure.md
Publishing	music-label/docs/echo-publishing.md
Open Platform	music-label/docs/open-platform.md
Label Partnerships	music-label/docs/label-partnerships.md
Backend Architecture	music-label/docs/backend-architecture.md
No-Code Build Guide	music-label/docs/no-code-build-guide.md
Tech Stack	music-label/docs/tech-stack.md
Tool Audit	music-label/docs/tool-audit.md
Policies & Standards	music-label/docs/policies-and-standards.md
New Ideas	music-label/docs/new-ideas-v1.md
Agent Review	music-label/docs/agent-review-v17.md
Phase 3 Pages	music-label/docs/phase3-bubble-pages.md
Phase 5 Agents	music-label/docs/phase5-make-agents.md
Presentation (20 slides)	music-label/presentation/
Branding Checklist	music-label/ECHO-Branding-Checklist.pdf

11. INFRASTRUCTURE

Component	Details
GitHub Repo	github.com/GKOffice/echo-music-company
Bubble Prototype	jeff-35035.bubbleapps.io (still live)
Airtable Base	appO0GzjvAcH1qEsd (7 tables)
Google Drive Backup	Melodio-Backup-2026-03-16/ (code zip + session PDF)
Platform Overview PDF	Desktop/Melodio-Platform-Overview.pdf (3.8MB)
Docker Compose	PostgreSQL + Redis + API + Web + Agents
Local Stack	PostgreSQL :5432, Redis :6379, API :8000, Web :3000

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12. CRITICAL BUSINESS RULES

- **NEVER say "invest/investment/ROI/returns"** → use "buy/purchase/own/earn"
- **No long-term contracts EVER** — per-song or per-album deals only
- **80% Marketing Rule** — artist point sales: 80% → marketing, 20% → pocket
- **Artist keeps 100% publishing** — Melodio takes 10% admin fee only
- **Points = product sale (license agreement), NOT securities**
- Master reversion: 5yr or 3x recoup, 15% perpetual override
- Only recording costs recoup (not marketing, PR, video, distribution)
- Artist minimum 30 master points retained per track
- Producer points from label's share always
- Ambassador program is invite-only, NOT publicly advertised
- All artist communication exclusively via Comms Agent (#14)

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13. FINANCIAL PROJECTIONS

Year	Revenue
Year 1	$448K

Year	Revenue
Year 2	$2.73M
Year 3	$9.89M
Year 5	$25-50M

MVP Cost: $884 upfront, ~$80-170/mo ongoing

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14. WHAT'S STILL NEEDED

Priority	Item	Est. Cost
🔴	Securities attorney opinion letter	$5-15K
🔴	Terms of Service (attorney review)	$2-5K
🔴	10 contract templates (attorney drafts)	Included
🟡	Deploy to melodio.io (Railway/Vercel)	~$20-50/mo
🟡	Real auth system (functional login)	Dev time
🟡	Stripe webhook secret	Free (dashboard)
🟡	Spotify + YouTube API keys	Free
🟡	DocuSign integration	~$10-25/mo
🟢	First artist signed	—
🟢	First release	—

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15. GIT HISTORY (22 Commits)

```
719496d feat: Artist Intelligence — search, AI analysis, 10 predictive signals
682d9f5 fix: handle sources_unavailable as objects
4519d92 fix: handle API response shapes
0a2c4e2 feat: Fan dashboard, artist profiles, ambassador, transparency, etc.
8a67b3f feat: Wire pages to API, search/filter, mobile, onboarding wizard
33ee1bc feat: Fan Intelligence Agent #23
369cf5c feat: Stripe payment integration
134946f feat: Digital Merchandise system
```

```
fb1cd84 feat: Legal Agent — worldwide compliance
fc51e7e feat: Songwriter Portal
4523d39 feat: Deal Room extensions
8b56366 feat: Creator Deal Room
060d4f3 rebrand: ECHO → Melodio
1a08277 feat: Social, Marketing, Creative, Sync, Artist Dev, PR, Comms
agents
612db47 feat: QC, Infrastructure, Intake, Merch, YouTube, Hub agents
827d694 feat: ECHO Points Vault — Store, Exchange, pricing, payouts
8eee3d4 feat: Analytics + Distribution agents
24d85fd feat: Finance + Legal agents
b2269dc feat: Artist dashboard, Points Store, landing, demo submission
0433ed6 feat: Phase 3 foundation — monorepo, DB, FastAPI, Next.js 14
138587c init: ECHO project
```

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